

With machine learning, can we abolish experimental chemistry?

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At first look this question appears to paint experiment as a cost to progression in chemistry, portraying machine learning as a cheaper and quicker alternative, and on the surface, this seems, to an extent, fairly correct: machine learning can threaten experimental chemistry as it can find and extract patterns from vast amounts of data, making experiments look slow and expensive in comparison. Reversing what has long been the norm of experiment as a gold standard and computer calculations as an approximation. However, by saying this, we are fundamentally assuming that experiments mainly exist to generate the data or examples to build theory, but in reality, they can act more importantly: they test theories and computations against reality, especially useful in uncertain or new cases. Therefore, rather than saying machine learning should fully eliminate the need for experiments, it is better to instead note the change in their role in scientific progression, making them more selective, strategically chosen and keeping them necessary for when the predictions made by machine learning may be unreliable. To ensure their coexistence is most optimal, their individual strengths should be leveraged to act together, combining machine learning's power to make accurate predictions in chemistry's huge search spaces with experiments that turn prediction into discovery and explanation.

Machine learning does seriously pose a threat to replacing parts of experimental chemistry, a subject that often involves enormous search spaces, expensive testing and repeated trial and error, all while facing pressures to keep costs low. Chemists rarely choose between only a few options; instead, they must calibrate across all the possible molecules, materials, reaction conditions, catalysts, etc., where even the slightest changes can produce different outcomes. Thus, the main difficulty is finding where exactly it is worthwhile to experiment. Furthermore, the problem is made more prominent by the fact that each experiment has a cost, such as time, materials and labour, such that in practice you have limited resources to cover a huge range of possibilities. From here, making chemical advancements begins to look more like finding optimisations than developing theories, exactly where machine learning can naturally be incredibly useful.

This is especially clear in certain branches of chemistry, such as reaction optimisation, an area with real practical value, where optimising by adjusting conditions can be invaluable to industry and research. Here, the number of combinations of variables — temperature, concentration, solvent, catalyst and so on — grows far too quickly for exhaustive testing, so the problem becomes one of sequential search under limited resources, exactly where machine learning can be very effectively employed. By using previous results to estimate where better outcomes are likely, it becomes possible to choose the next experiment worth running rather than wasting resources on blind trial and error. This is not just a matter of picking the obvious option, but of balancing what already looks promising with more uncertain areas that may prove even better. This is essentially what Bayesian optimisation does: it guides decision-making under uncertainty, and Shields et al. report that in reaction-optimisation tasks it outperformed human decision-making in both average optimisation

efficiency, measured by the number of experiments, and consistency, measured by the variance of outcome against initially available data.

There is a core problem when we are using machine learning as an aid to chemical discovery: it is at its most powerful when the environment it is dealing with is still close enough to the environment it has already seen. However, the most valuable chemical research is often, by definition, not familiar and instead concerns new molecules, new materials, and unusual property values. Therefore, we are relying on a model trained on past chemical data to perform on new cases, but the hardest cases are not simply new examples of familiar patterns and instead include possibilities beyond the range of the training datasets. Recent work makes this especially clear. Segal et al. argue that discovering high-performing materials and molecules often requires identifying extremes with property values outside the known distribution, while Shimakawa et al. emphasise the inherent difficulty of extrapolative prediction beyond the training distribution in molecular property prediction, particularly when the data are limited. In cases like these, machine learning's apparent confidence can become misleading: a model may still return a clear and precise-looking prediction even when it has moved beyond the region in which it has been properly trained.

It is in scenarios like these, where candidates for certain tests are far from what the model has been trained on, that experiment becomes vital as a means to test epistemically. Work in npj Computational Materials by Schultz et al. shows that knowing a model's "domain of applicability" - the set of molecules that are similar enough to the training data that the model's predictions are likely to be reliable — is essential for obtaining reliable predictions. They also found that the greater the "dissimilarity" of a new data point from the training data, the worse the model's performance became, and that this degradation can cause not only errors in the prediction, but also the uncertainty estimate itself. If the domain of the machine learning model is limited, then we simply cannot trust the results it provides, and this matters most in chemistry, where discovery is most often unfamiliar. As a result, we must turn to experiment to definitively check whether a new finding is truly real rather than just statistically likely based on previous results. While machine learning can narrow and prioritise experiments, it cannot totally abolish them. As machine learning improves at finding these results, it becomes more important to know the difference between its hallucinations and the true results.

It is important that we make the distinction that experiment isn't just a fallback for when machine learning fails to provide accurate results; it is the primary process in which science can gain new ground and eliminate uncertainties. By conducting experiments, we are not just trying to add rows to datasets but instead obtaining a greater understanding of how things act in the real world that machine learning models do not already possess. Therefore, when experiments become expensive or limited, abolishing experiments doesn't come into question; it's a question of using ML to find the specific experiments that should be carried out. Machine learning doesn't then remove the need to test in reality, but alternatively can decide where testing reality will teach us the most. In turn, this in fact contributes to making experiments more important, as, although fewer tests need to be run, the ones that do can be decisive and greatly extend knowledge.

In practice, this is precisely what's happening. Some of the most cutting-edge and modern labs that are at the forefront of scientific and chemical discovery aren't purely run on machine

learning prediction models, and still heavily rely on real-world experiments. A prime example is A-lab, an autonomous lab for material synthesis, which has managed to combine machine learning trained on historical literature and diffraction data, with robotics to plan and interpret experiments, synthesising 36 out of 57 target materials over just 17 days. It was built to speed up and optimise the experimental realisation of materials, but fundamentally, that did not mean getting rid of it altogether; it ran experiments, interpreted results and proposed improved follow-ups. Thus, showing that even the most advanced systems still depend on real experiments inside their testing loops, and that the chemical discoveries work best in a hybrid between machine learning and experiments.

Until now, we have mainly focused on the accuracy of machine learning being the primary reason behind its inferiority to experiments, but there lies a further issue, even when it predicts with utmost accuracy, that being how it actually contributes to scientific knowledge. Prediction is not interchangeable with explanation, and chemists are not solely interested in outcomes but more in understanding the causes of said outcomes, its mechanisms and why it occurs. This is something that we cannot achieve with a machine learning model alone, as, although it may output highly precise forecasts, those forecasts are still, in themselves, statistical outputs from the model: they may show that a pattern holds, without showing what underlying mechanism makes it hold. Genuine scientific explanations hold more value to chemists than a successful prediction because they still uphold when you tweak different variables, whereas by following patterns, a certain idea may only work when nothing is changed. By experimenting, you are probing why things happen by changing conditions and isolating variables to get a solid, holistic understanding of the mechanism, something machine-learning models alone are not capable of providing.

With this in mind, the optimal approach for obtaining the best results is by incorporating chemical knowledge with machine learning, moving away from the model just giving answers. Many recent models do this by embedding domain knowledge and becoming more “chemistry-aware” rather than relying on pattern recognition alone, as seen in a 2023 Nature Communications paper on reaction performance prediction using a knowledge-based graph model. In the paper, models are trained on steric and electronic information in order to help them explain the predictions and make more accurate predictions beyond their training data. Yet, importantly, they still verified these predictions with real experiments to gain concrete evidence of their validity, so even the best response to ‘black box’ models still ends in experiments, proving once again that even the most chemistry-aware machine learning does not abolish experiment; it simply changes experiment from broad testing into final verification of whether a model’s explanation is actually true.

In certain data-rich and optimisation-heavy areas of chemistry, the title is more plausible than what earlier sections may suggest. Tasks that involve repetition and optimisation of well-understood things, machine learning can automate much of the experimental process, especially with the rise of self-driving labs. Many sources support this outlook, with a 2024 paper on self-driving labs notably stating, “autonomous experimentation can navigate complex and exponentially expanding reaction spaces with an efficiency unachievable through conventional experimentation”. However, one could argue that this is simply abolishing the human-performed iterations and not the experiment as a whole. The distinction between the abolition of experimental labour and the abolition of experimental science is where this argument falls down, as the system does still rely on physical interventions and

real-world validations, no matter how autonomous. So, although parts of experimental labour may be replaced, experimentation as a scientific method still remains.

Another version of this argument can be seen in certain highly data-heavy scientific applications such as AlphaFold, where machine learning comes remarkably close to abolishing some experimental areas altogether. Yet even here, its success still depends on prior experimental data, benchmarking, and ongoing validation. So, while machine learning may dramatically reduce the volume of experimentation required, even one of the clearest modern examples of applied machine learning has not fully abolished the need for experimentation.

Ultimately, whether machine learning can abolish experimental chemistry depends on what is meant by “abolish”. On one hand, it is becoming increasingly true with machine learning and its self-driving labs already removing the need for repetitive screenings and trial and error. However, this is not an abolishing experiment in the pure scientific sense. As mentioned, experimental chemistry is not just a means to generate data that machine learning might one day replace. It is instead a way to test predictions against reality, to eliminate uncertainties and derive a deeper understanding as opposed to surface-level pattern recognition. Although models rank statistical candidates, optimise conditions, and can even appear to perform beyond their training data, they cannot, by themselves, fully confirm that a new finding truly behaves as predicted or that a successful pattern allows for deeper understanding. Therefore, the future of chemistry does not see experiments disappear, but instead sees a shift in its role. As machine learning compresses searches, sharpens choices, and automates much of human labour, experiments will remain the final, decisive act whereby chemistry can gain new knowledge. What machine learning can abolish is not experimental chemistry itself, but only its least intelligent forms.

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